

SQA & Testing

Chapter 2

Software Testing

Types of software testing

Static Testing Vs Dynamic Testing

Static Testing:

SW is tested without code execution. Manual or automated review of code, requirement document and design are done in order to find errors. The main objective is to improve quality of SW by finding errors in early stage of software development.

Dynamic Testing:

Here, a code is executed. The main objective is to confirm that the software product works in conformance with the business requirement.

It needs to execute the SW and validate the output with the expected outcome.

It is performed at all **levels of testing** and it can be either **black or white box testing**.

Simply, Static testing is about prevention of defects whereas Dynamic testing is about finding and fixing the defects.

...Types of software testing

Static Testing

Review

Refers to informal check or inspection of technical documents produced by someone else, either organized as individual effort, or as group effort in meetings, conference calls, etc.

Focuses on logical and conceptual problems.

Walkthrough

A more organized review typically applied to software design and code. Meetings are usually used for these walkthroughs.

The designer or the code owner usually leads the meeting, explaining the intentions and rationales for the design/ code, and the other reviewers (meeting participants) examine these design/ code for overall logical and environmental soundness and offer their feedback and suggestions. (Defect detection is not the focus.)

Software inspection

Deals with software defects already injected into the software system by detecting their presence through critical examination by human inspectors.

(steps: Planning and preparation -> Inspection or collection -> Correction and follow-up)

...Types of software testing

Dynamic Testing

Functional Testing:

Verifies that each function of SW application operates in conformance with the requirement.

Every functionality of the system is tested by providing appropriate input, verifying the output and comparing the actual results with the expected results.

Non-Functional Testing:

To check non-functional aspects (performance, useability, scalability, load, reliability, stress, compliance, portability, disaster recovery, volume...) of SW.

Hard to test using manual testing.

...Types of software testing

White-box & Black-box testing

Based on the sources of information for test design, testing has two broad concepts: white-box (structural testing techniques) and black-box testing (functional testing techniques).

...Types of software testing

White-box testing

Examining of source code with a focus on control flow and data flow.

Control flow refers to flow of control from one instruction to another. Control passes from one instruction to another instruction in a number of ways, such as one instruction appearing after another, function call, message passing, and interrupts. Conditional statements alter the normal, sequential flow of control in a program.

Data flow refers to the propagation of values from one variable or constant to another variable. Definitions and uses of variables determine the data flow aspect in a program.

...Types of software testing

Black-box testing

One does not have access to the internal details of a program and the program is treated as a black box.

A test engineer is concerned only with the part that is accessible outside the program, that is, just the input and the externally visible outcome.

A test engineer applies input to a program, observes the externally visible outcome of the program, and determines whether or not the program outcome is the expected outcome.

Inputs are selected from the program's requirements specification and properties of the program's input and output domains.

A test engineer is concerned only with the functionality and the features found in the program's specification.

TEST LEVELS

Testing is performed at different levels involving the complete system or parts of it throughout the life cycle of a software product.

A software system goes through four stages of testing before it is actually deployed. These four stages are known as

- Unit,
- Integration,
- System, and
- Acceptance level testing.

The first three levels of testing are performed by a number of different stakeholders in the development organization, where as acceptance testing is performed by the customers.

... TEST LEVELS

Unit Test (COMPONENT LEVEL TEST)

Individual components are tested independently to ensure their quality.

The focus is to uncover errors in design and implementation, including

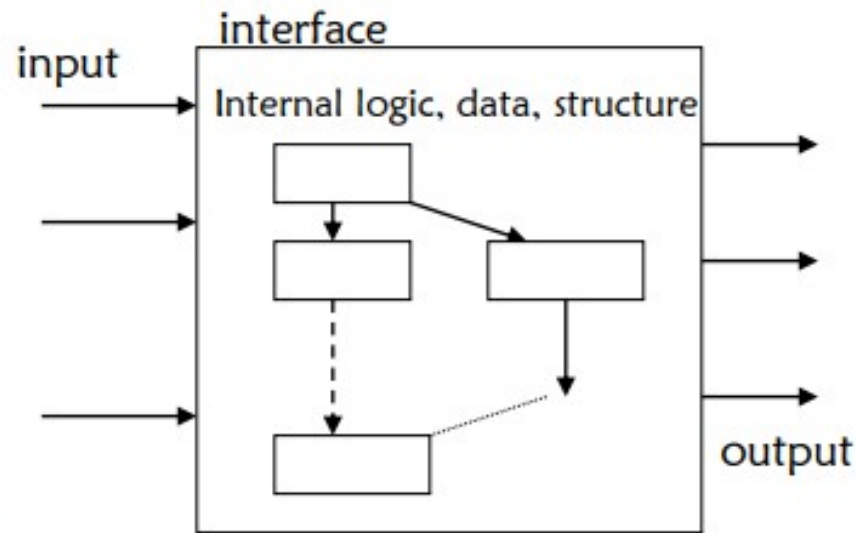
- Data structure in a component
- Program logic and program structure in a component
- Component interface
- Functions and operations of a component

Unit testers: developers of the components

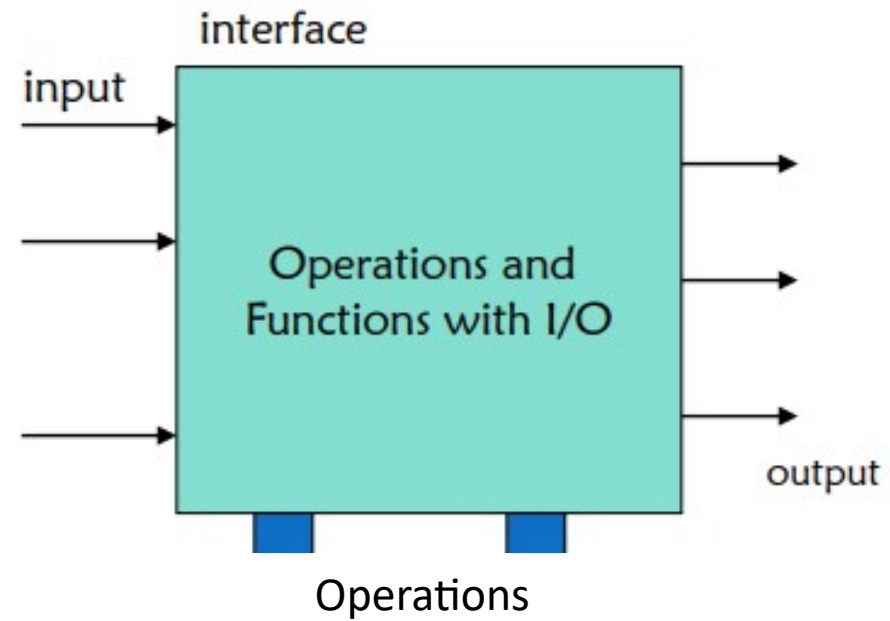
... TEST LEVELS

...Unit Test

White-box



Black-box



... TEST LEVELS

Integration test

A group of dependent components are tested together to ensure their the quality of their integration unit

The focus is to uncover errors in:

- Design and construction of software architecture

- Integrated functions or operations at sub-system level

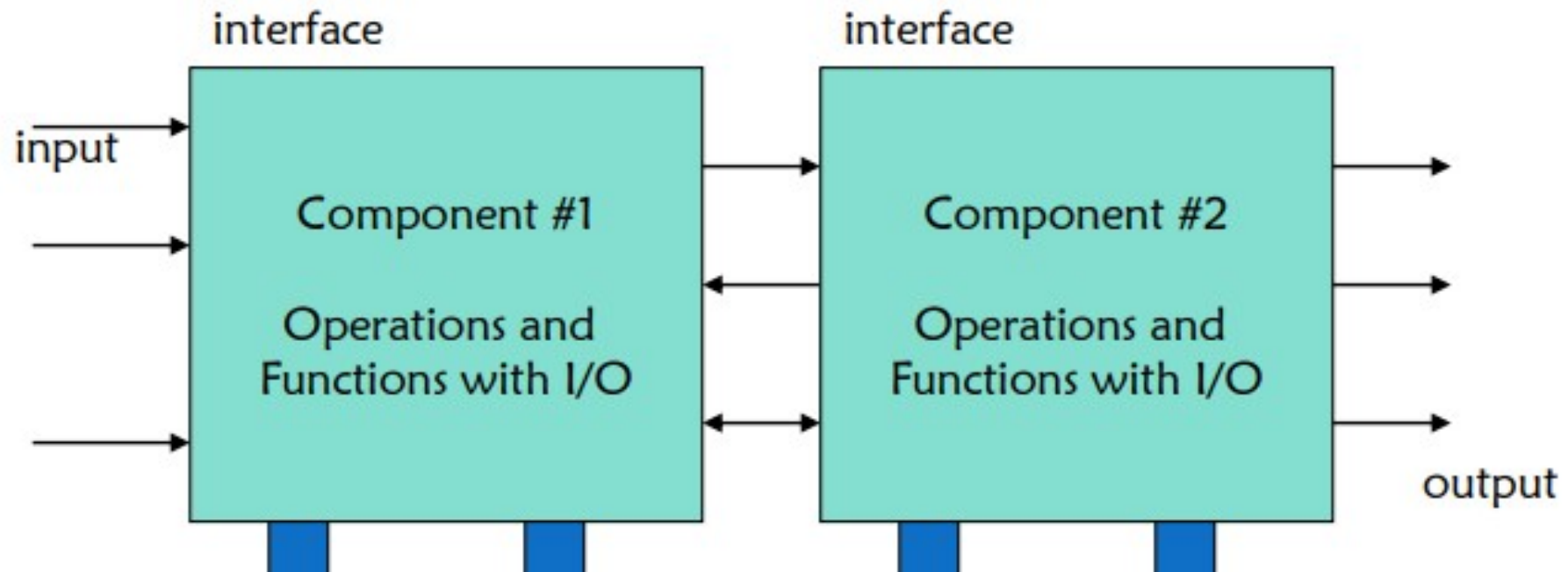
- Interfaces and interactions between them

- Resource integration and/or environment integration

Integration testers: either developers and/or test engineers

... TEST LEVELS

... Integration test



... TEST LEVELS

System testing

The system software is tested as a whole

It verifies all elements mesh properly to make sure that all system functions and performance are achieved in the target environment

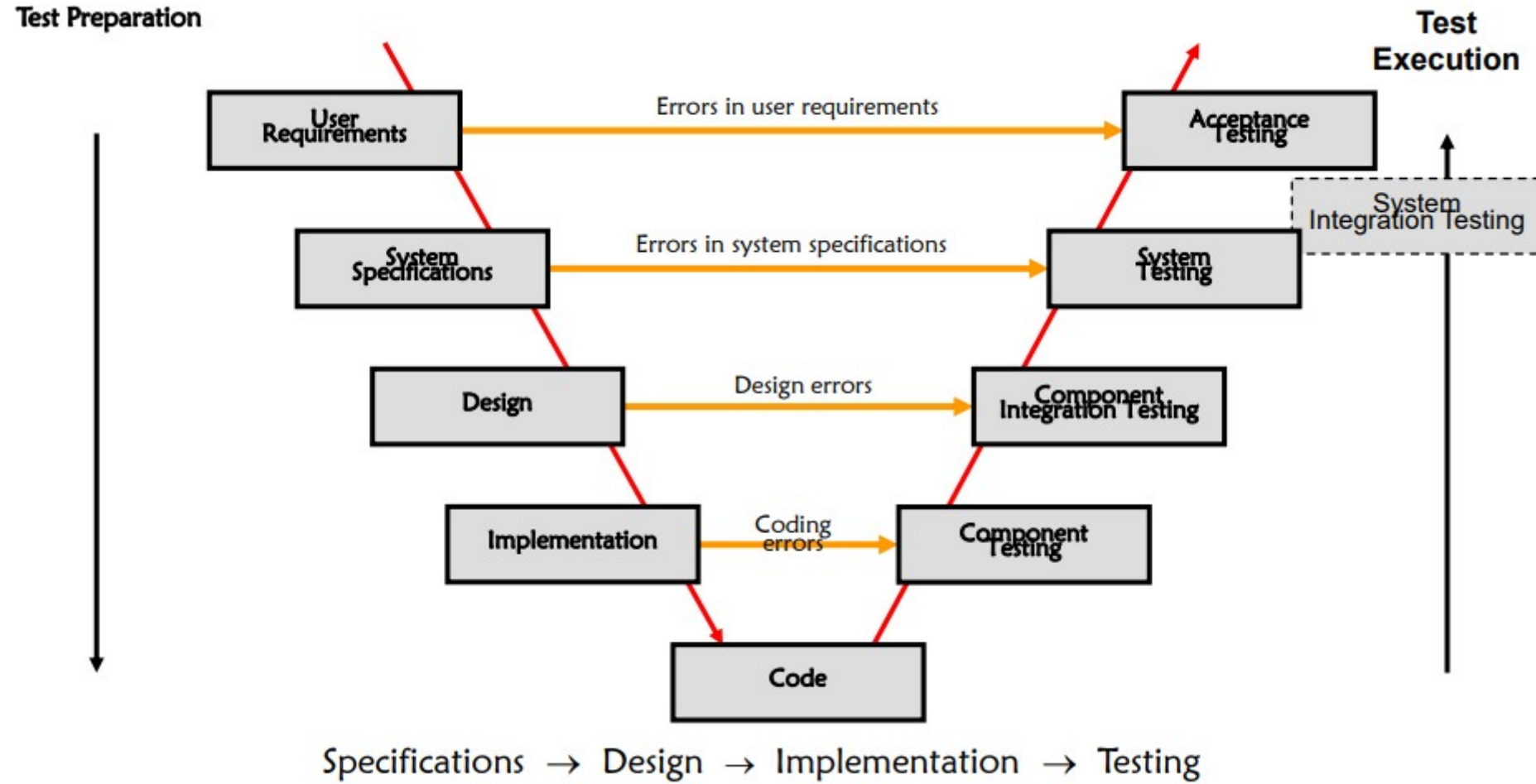
The focus areas are:

- System reliability and recoverability (recovery test)
- System installation (installation test)
- System behavior in the special conditions(stress and load test)
- System user operations (acceptance test/alpha test)
- Hardware and software integration and collaboration
- Integration of external software and the system

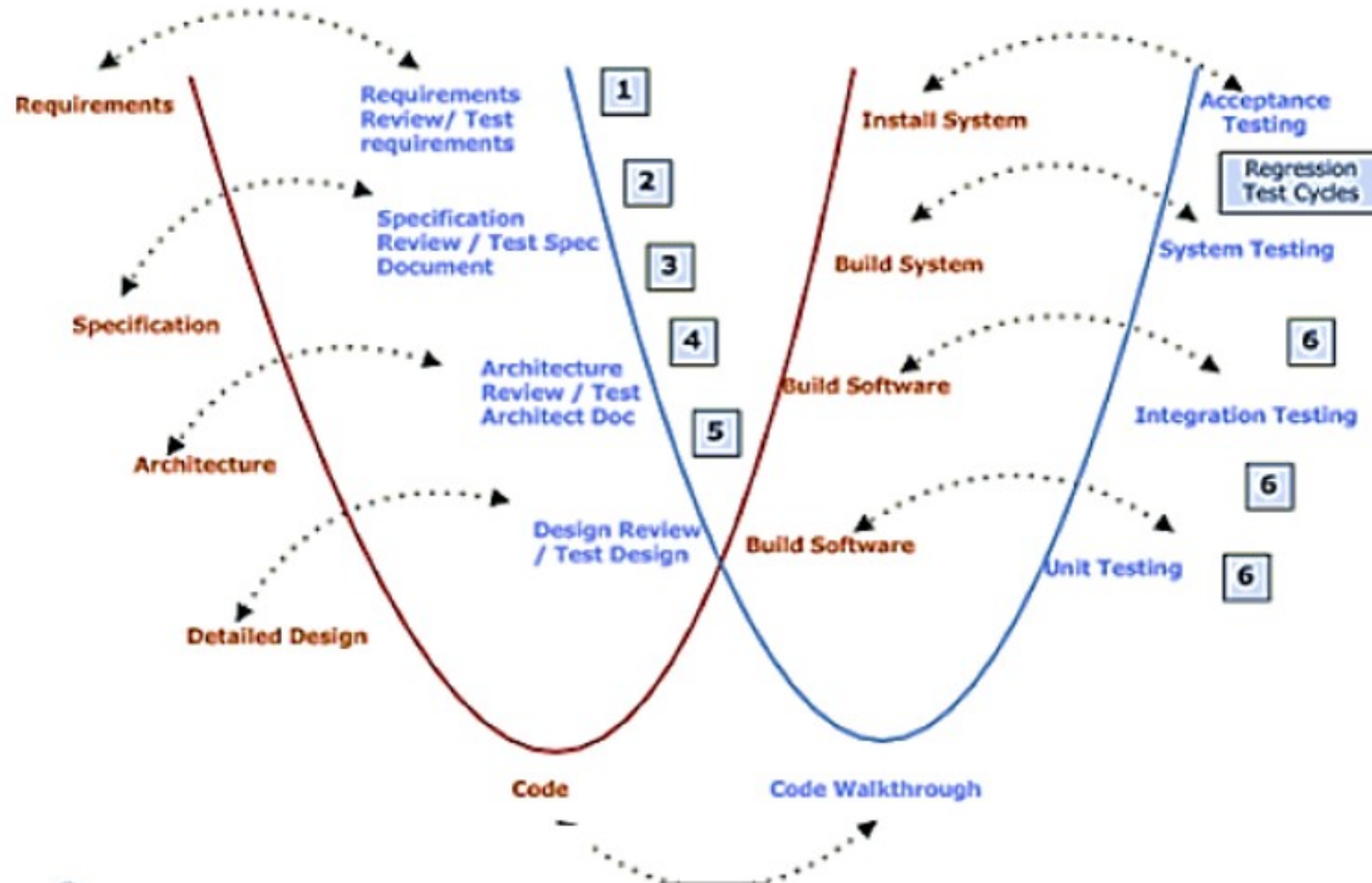
System testers: test engineers in ITG or SQA people

When a system is to be marketed as a software product, a testing process called beta testing is often used

Models for testing (V-model)



Models for testing (W-model)



Alpha and Beta Testing

Alpha site and beta site tests are employed to obtain comments about quality from the package's potential users. They are additional commonly used tools to identify software design and code errors in software packages in commercial over-the-counter sale.

... Alpha and Beta Testing

Alpha site tests are tests of a new software package that are performed at the developer's site.

The customer, by applying the new software to the specific requirements of his organization, tends to examine the package from angles not expected by the testing team.

The errors identified by alpha site tests are expected to include the errors that only use by a real user can reveal, and thus should be reported to the developer.

... Alpha and Beta Testing

Beta site tests are much more commonly applied than are alpha site tests.

The beta site test process can be described as follows. Once an advanced version of the software package is available, the developer offers it free of charge to one or more potential users.

The users install the package in their sites (usually called the “beta sites”), with the understanding that they will inform the developer of all the errors revealed during trials or regular usage.

Participants in beta site testing are often users of previously released packages, sophisticated software professionals and the like.

RE-TESTING AND REGRESSION TESTING

Re-testing: re-running of test cases that caused failures during previous executions, after the (supposed) cause of failure (i.e. fault) has been fixed, to ensure that it really has been removed successfully.

Regression testing: re-running of test cases that did NOT cause failures during previous execution(s), to ensure that they still do not fail for a new system version or configuration.

Debugging: the process of identifying the cause of failure; finding and removing the fault that caused failure.

TYPES OF ACCEPTANCE TESTING

- A product is ready to be delivered to the customer after the system test group is satisfied with the product by performing system-level tests.
- Customers execute acceptance tests based on their expectations from the product.
- The customer generally reserves the right to refuse to take delivery of the product if the acceptance test cases do not pass. There are two categories of acceptance testing:
 - User acceptance testing.
 - Business acceptance testing

...TYPES OF ACCEPTANCE TESTING

- The UAT is conducted by the customer to ensure that system satisfies the contractual acceptance criteria before being signed off as meeting user needs.
- The BAT is undertaken within the development organization of the supplier to ensure that the system will eventually pass the UAT. It is a rehearsal of UAT at the premises of the supplier.
- The development organization of the supplier derives and executes test cases from the client's contract, which include the acceptance criteria